

Quiz 2 Practice Problems

10-606

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1 Matrix Inverses

1. True/False: If $AB = I$, then $BA = I$. **Solution:** True. If $AB = I$, then A and B are inverses of each other, so $BA = I$ as well.
2. Prove or disprove: Every square diagonal matrix is invertible. **Solution:** False. A diagonal matrix is invertible iff all diagonal entries are nonzero.
3. Compute the inverse of $\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$. **Solution:** $\det = 2 \cdot 3 - 5 \cdot 1 = 1$.
Inverse $= \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$.
4. Suppose A is invertible. Show that $(A^T)^{-1} = (A^{-1})^T$. **Solution:** $(A^T)(A^{-1})^T = (A^{-1}A)^T = I^T = I$. Thus $(A^{-1})^T$ is the inverse of A^T .
5. Explain, without resorting to calculation, why a matrix with two identical rows cannot be invertible. **Solution:** Identical rows $\implies$ linearly dependent $\implies$ determinant $= 0$, so not invertible.
6. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{bmatrix}$. Is A invertible? **Solution:** No. Last row is zero $\implies \text{rank} < 3 \implies$ singular.
7. If A is invertible, does $Ax = b$ have a unique solution for all b ? **Solution:** Yes. Solution is $x = A^{-1}b$. Existence and uniqueness follow from invertibility.
8. If $Ax = b$ has no solution, show that $ABx = b$ has no solution for any matrix B . **Solution:** Suppose it did, then $z = Bx$ satisfies $Az = b$. But such a z cannot exist by assumption.

9. A set of vectors $\{v_1, \dots, v_n\}$ in $\mathbb{R}^n$ are *orthonormal* if $v_i^\top v_j = 1$ if $i = j$ and 0 otherwise. A matrix $Q \in \mathbb{R}^{n \times n}$ is orthonormal if its column and row vectors are orthonormal. If Q is orthonormal, show that $Q^\top = Q^{-1}$. **Solution:** Let $Q = [v_1, \dots, v_n]$ where $\{v_1, \dots, v_n\}$ are orthonormal. Then $(Q^\top Q)_{ij} = v_i^\top v_j = \delta_{ij}$. That is, $Q^\top Q = I$, implying that $Q^\top$ is the inverse of Q .

Linear Systems

1. Can the following system be solved? $\{x + y = 2, 2x + 2y = 5\}$. **Solution:** Inconsistent: first eq $\implies 2x + 2y = 4$, but second requires 5. No solution.
2. How many solutions are there to $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} x = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$. **Solution:** Row 2 is multiple of row 1. Solutions: $x_1 + 2x_2 = 3 \implies x_1 = 3 - 2x_2$. Infinitely many.
3. Row-reduce $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$. **Solution:** Final echelon form: $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$.
4. For which a is $\begin{bmatrix} 1 & a \\ 2 & 4 \end{bmatrix}$ invertible? **Solution:** $\det = 4 - 2a$. Invertible iff $a \neq 2$.
5. Solve the system: $\{x + y + z = 6, x - y + z = 2, 2x + z = 5\}$. **Solution:** $x = 1, y = 2, z = 3$.
6. True/False: If the equation $Ax = 0$ has only the trivial solution $x = 0$, then the columns of A form a basis for $\mathbb{R}^n$. **Solution:** True. Trivial solution means full column rank $= n$. Columns are independent and span $\mathbb{R}^n$.
7. Solve $Ax = b$ where $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, b = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$. **Solution:** You could row-reduce, or notice that $\det A = -2$. Hence $A^{-1} = \frac{1}{-2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$. Multiplying by b gives $x = (1, 2)^\top$.
8. Construct 3 equations in 3 unknowns with no solutions. **Solution:** Example: $x + y + z = 1, 2x + 2y + 2z = 2, x + y + z = 3$. First and third inconsistent.

Eigenvalues and Eigenvectors

- Find the characteristic polynomial of $A = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$ for any $a \in \mathbb{R}$ (i.e., compute $\det(A - \lambda I) = 0$). **Solution:** $(1 - \lambda)^2$. Only eigenvalue=1, algebraic multiplicity 2.
- What sort of geometric action does the matrix $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ correspond to? **Solution:** 90 degree rotation CCW.
- What sort of geometric action does the matrix $\begin{bmatrix} -3 & 0 \\ 0 & 3 \end{bmatrix}$ correspond to? **Solution:** Flips the vector along the x-axis; scales it by 3.
- Prove that an orthogonal matrix (see Section 1) always has eigenvalues of magnitude 1. **Solution:** We showed above that $Q^T Q = I$. Let x be eigenvector of Q with eigenvalue λ , so $Qx = \lambda x$. Multiplying both sides by, $(Qx)^T$:

$$x^T Q^T Q x = (x^T Q^T) \lambda x = \lambda x^T Q x = \lambda^2 x^T x.$$

The left hand side is equal to $x^T x$. Hence $x^T x = \lambda^2 x^T x$, implying that $\lambda^2 = 1$.

- If 0 is an eigenvalue of A , what does this tell you about the rank of A ? **Solution:** A is not full rank (its kernel has dimension at least 1).
- Let A have eigenvectors v_1, v_2 with eigenvalues λ_1, λ_2 . Suppose $x = 3v_1 - 2v_2$. What is Ax in terms of v_1, v_2 ? Explain geometrically what happened to x .

Solution: Since $Av_1 = \lambda_1 v_1$ and $Av_2 = \lambda_2 v_2$, $Ax = 3\lambda_1 v_1 - 2\lambda_2 v_2$. Geometrically, x is decomposed into eigen-directions. Each component is scaled by its eigenvalue: stretched/compressed (if $|\lambda_i| \neq 1$), flipped (if $\lambda_i < 0$), or left unchanged (if $\lambda_i = 1$).